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A Report on

Memory Optimization for NPC (Non-Playable Characters) AI for Long Term Behaviour in Video Games

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Abstract

With the rise of increasingly immersive and complex video games, the demand for intelligent, adaptive Non-Playable Characters (NPCs) has never been greater. Current AI systems rely on pre-scripted behaviors and finite state machines, resulting in static, predictable interactions that lack long-term memory and break player immersion. CSAM (Cognitive Sparse Access Memory) is a novel AI architecture designed to address these fundamental limitations. It synthesizes state-of-the-art research in cognitive AI, scalable machine learning, and intelligent information retrieval to create a complete, end-to-end memory solution for lifelike agents.

The CSAM platform utilizes a cognitively-inspired, three-tiered memory system to store and process an agent's experiences. By leveraging Sparse Access Memory (SAM) for O(log N) retrieval and a Maximal Marginal Relevance (MMR) algorithm for ensuring cognitive diversity, CSAM enables NPCs to form long-term memories, develop evolving relationships, and exhibit nuanced, non-repetitive behaviour in real-time. This project details the architecture of CSAM, validates its novelty against existing research, and provides a clear plan for its implementation and evaluation, paving the way for the next generation of intelligent virtual characters

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1. Introduction

1.1 Overview

The video game industry has achieved near-photorealism in visual presentation, yet the behavioral intelligence of Non-Playable Characters (NPCs) remains rudimentary. Most NPCs operate as complex puppets, following pre-scripted logic that renders them incapable of genuine memory, learning, or adaptation. This creates a jarring disconnect between how characters look and how they act, ultimately limiting narrative depth and player immersion. Current systems, bound by technical constraints, cannot support the kind of persistent, long-term memory that is the hallmark of believable intelligence. This project introduces CSAM, a novel memory architecture designed to bridge this gap.

1.2 Motivation

The motivation for CSAM stems from the desire to create truly dynamic and evolving virtual worlds. Players today expect characters to remember their actions, to form relationships, and for the game world to react to their unique journey. An NPC that treats the player like a stranger after a shared, critical story moment is a fundamentally broken experience. By solving the core technical challenges of memory scalability and intelligent retrieval, we can unlock new forms of interactive storytelling and create characters that feel less like code and more like companions, rivals, and living inhabitants of a persistent world.

1.3 Problem Statement and Objectives

Problem Statement:

Current AI memory solutions for interactive agents are incapable of supporting coherent, long-term behavior in real-time due to a combination of cognitive immaturity, prohibitive computational costs, and simplistic retrieval mechanisms.

Objectives:

The aim of this project is to design, implement, and validate the CSAM architecture to address this problem. The primary objectives are to:

1. Create a cognitively-inspired memory architecture with distinct short-term, episodic, and consolidated long-term memory layers that mimic human-like memory processes.
2. Ensure real-time performance by integrating a Sparse Access Memory (SAM) retrieval engine capable of O(log N) search complexity, allowing the system to scale to a massive memory archive without performance degradation.
3. Implement an intelligent retrieval mechanism that prioritizes not just relevance but also cognitive diversity, using a Maximal Marginal Relevance (MMR) algorithm to prevent behavioral repetition.
4. Develop a proof-of-concept in the Godot game engine to empirically demonstrate the qualitative (believability) and quantitative (performance) benefits of the CSAM architecture.

1.4 Organization of the Report

This report is structured into several chapters. Chapter 2 provides a detailed survey of the foundational literature and identifies the critical research gaps our project addresses. Chapter 3 presents the proposed CSAM system, detailing its architecture, methodology, and technical requirements. Chapter 4 discusses the data requirements for training and evaluation, and Chapter 5 outlines the project work plan for the upcoming semester.

2. Literature Survey

2.1 Survey of Foundational Research

Following table summarizes the foundational papers that inform our approach.

Author & Year	Title / Source	Methodology	Key Findings & Contributions	Limitations Addressed by Our Work
Zheng et al., 2024	"MemoryRepository for AI NPC" - IEEE Access	Implements a hierarchical memory with distinct Short-Term and Long-Term "rooms" for LLM-based agents.	Provides a cognitive framework that enables more human-like conversations through memory summarization.	Does not address the critical performance bottleneck of searching a large memory archive in real-time.
Rae et al., 2016	"Scaling Memory-Augmented Neural Networks with Sparse Reads and Writes" - NeurIPS	Introduces Sparse Access Memory (SAM), which uses Approximate Nearest Neighbor (ANN) indexes to achieve O(log N) memory retrieval time.	Proves that massive performance gains are possible without sacrificing learning efficiency, solving the scalability problem.	Provides a purely technical performance solution but does not address the cognitive structure or the quality of the retrieved memories.
Phatak et al., 2025	"Inducing Diversity in Differentiable Search Indexing" - arXiv	Integrates a Maximal Marginal Relevance (MMR) inspired component directly into the training process of a	Demonstrates that a model can learn to intrinsically retrieve a set of results that is both relevant and diverse,	The scope is limited to information retrieval and does not address the challenges of a dynamic, evolving memory store

		retrieval model.	preventing redundancy.	for an interactive agent.
Park et al., 2023	"Generative Agents: Interactive Simulacra of Human Behaviour" - UIST	Agents use a persistent memory stream and a "reflection" mechanism to synthesize high-level insights from raw experiences.	Successfully created believable, emergent social behaviours, validating the power of a persistent, human-like memory.	The architecture was not designed for real-time performance and incurred significant latency, making it unsuitable for interactive games.

Cognitive Architecture: The Human-like Blueprint

A significant advancement in creating human-like agents comes from Zheng et al. (2024), who proposed the MemoryRepository architecture[11]. This system implements a hierarchical memory with separate Short-Term and Long-Term "rooms" for LLM-based agents, using an LLM-driven summarization and forgetting mechanism to consolidate raw interactions into high-level memories[2222]. While this cognitive framework enables more believable, long-term conversations, its primary limitation is that it does not address the critical performance bottleneck of searching a large memory archive[333].

Scalable Performance: The Efficiency Engine

Addressing the performance issue is the work of Rae et al. (2016)on **Sparse Access Memory (SAM)**[4]. This research introduces a method to scale memory-augmented neural networks by using **Approximate Nearest Neighbor (ANN) indexes** to find relevant memories[5]. This

reduces the time complexity of a memory search from a linear scan (O(N)) to a logarithmic function:

Time Complexity=O(logN)

This equation signifies that even if the number of memories (N) increases a millionfold, the search time only increases by a small, constant factor. The paper proved that this method could be up to

1600x faster than dense memory models without sacrificing learning efficiency[6]. However, the work is a purely technical performance solution; it does not address the cognitive structure of the memory itself or the quality of the retrieved memories[7].

Intelligent Retrieval: The Diversity Curator

To enhance the quality of retrieval, the work of Phatak et al. (2025)on **Differentiable Search Indexing (DSI)** introduces a mechanism to ensure cognitive diversity[8]. They integrate a component inspired by**Maximal Marginal Relevance (MMR)** directly into the model's training loss function. The MMR algorithm seeks to maximize a combined score that balances relevance to the query with novelty from already selected results[9]:

$$MMR=D_i \in C \setminus S \operatorname{argmax} [\lambda \cdot Sim(Q,D_i) - (1-\lambda) \cdot D_j \in S \max Sim(D_i, D_j)]$$

This allows the model to learn to intrinsically retrieve a set of results that is both relevant and diverse, preventing redundancy without adding latency during inference[10]. However, its scope is limited to information retrieval and does not address the challenges of managing a dynamic memory store for an interactive AI agent[11].

Validation and The Remaining Gap

Finally, the landmark "Generative Agents" paper by Park et al. (2023) validates the power of a persistent, human-like memory[12]. Their agents used a memory stream and a "reflection" mechanism to synthesize high-level insights from raw experiences, leading to believable, emergent social behaviors[13]. Critically, the authors noted that their architecture was **not**

designed for real-time performance and incurred significant latency and cost, making it unsuitable for interactive games and highlighting the very performance gap that our work aims to solve[14].

2.2 Limitations of Existing System & Research Gaps

Our analysis of the literature reveals three fundamental gaps that have prevented the development of truly intelligent, long-term NPCs.

- The Cognitive Gap: Current systems often treat an agent's memory as a simple, chronological log of events. This is fundamentally different from human cognition, where we forget precise, low-level details while retaining high-level, abstracted summaries. This architectural flaw leads to NPCs with either perfect, robotic recall or simplistic, rule-based forgetting, neither of which is believable.
- The Retrieval Gap: Even a fast retrieval system is not necessarily an intelligent one. A search based purely on relevance will often return a set of highly similar, redundant memories. This leads to one-dimensional behaviour and repetitive dialogue.

3. Proposed System: CSAM

3.1 System Architecture and Flow

The CSAM architecture is a multi-layered, end-to-end system designed to address the identified gaps. The flow begins with a **Player Query**, which is processed by the central **CSAM Agent**. This agent's "mind" is the **Cognitive Sparse Access Memory Architecture**, which is composed of a **Cognitive Control Unit** and three distinct memory layers.

The Three Memory Layers:

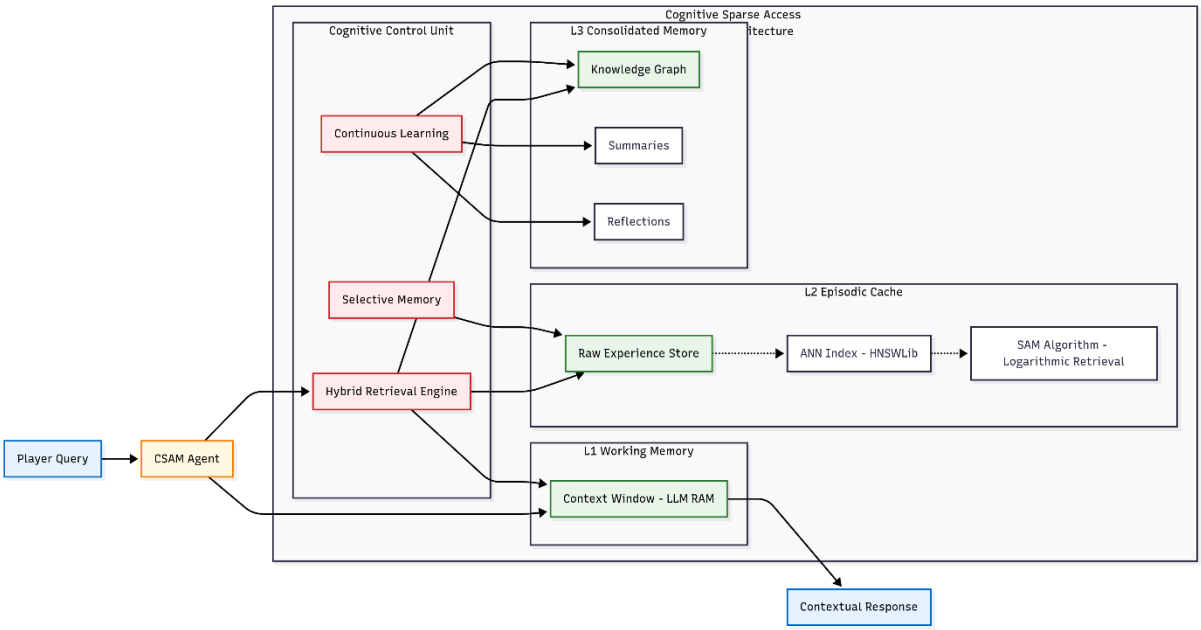
- **L1 Working Memory**: This is the agent's immediate consciousness, equivalent to an LLM's context window. It holds the current conversational context for generating immediate responses.
- **L2 Episodic Cache**: This layer stores the raw, unprocessed life experiences of the agent. It is a detailed log of events that serves as the source for long-term memory.
- **L3 Consolidated Memory**: This is the agent's deep, long-term memory. It does not store raw data, but rather structured knowledge in the form of **Summaries, Reflections,** and a **Knowledge Graph**.

The Cognitive Control Unit:

This is the active processing part of the brain.

- **Continuous Learning:** An asynchronous process that reads from the L2 cache and consolidates experiences into the L3 memory, effectively turning raw data into wisdom.
- **Selective Memory:** A process that prunes or de-prioritizes unimportant information in the L2 cache to manage memory cost and maintain relevance.
- **Hybrid Retrieval Engine:** The core of the agent's thought process. When queried, it pulls information from all three memory layers—L1 for context, L2 for specific facts, and L3 for deep insights—to form a comprehensive understanding before generating a response.

System Flow Diagram:



3.2 Methodology and Key Techniques

- **Cognitive Hierarchy:** The three-tiered memory system (L1, L2, L3) allows the agent to manage information at different levels of abstraction, enabling both detailed recall and high-level reasoning.
- **Sparse Access Memory (SAM):** The L2 Episodic Cache is indexed using a high-performance **Approximate Nearest Neighbor (ANN)** library (HNSWLib). This allows the retrieval engine to search a massive archive of raw experiences in **O(log N) time**, solving the performance bottleneck.
- **Maximal Marginal Relevance (MMR):** After the initial fast retrieval from the L2 cache, a secondary MMR algorithm re-ranks the candidate memories. This ensures the final set of memories passed to the LLM is not just relevant, but also diverse, preventing the agent from repeating itself.

3.3 Details of Hardware and Software Requirements

- **Hardware:**

- GPU-enabled server for model fine-tuning (e.g., NVIDIA RTX 3090 or higher).
- Development PC with minimum 16 GB RAM.

- **Software/Frameworks:**
  - **Game Engine:** Godot Engine 4.x
  - **Programming:** C++ (via GDExtension for core AI modules), GDScript (for high-level logic).
- **AI/ML Libraries:**
  - HNSWLib (for C++ ANN index).
  - A local LLM server (e.g., Ollama) for summarization and response generation.
  - Python with PyTorch/TensorFlow for any model fine-tuning.
- **Backend:** A simple Python server (e.g., FastAPI) may be used to host local AI models.

4. Relevant Dataset

This project's unique nature means there is no single, off-the-shelf dataset for training. The "data" is the dynamic stream of interactions generated during gameplay. Our data strategy is therefore twofold

1. **Dialogue and Personality Fine-Tuning Corpus:** To give our NPC a baseline personality and conversational ability, we will fine-tune a base LLM on a curated corpus of text. This will consist of open-source dialogue datasets from sources like movie scripts, literature, or existing games that match the desired character archetype.
2. **Synthetic Interaction Corpus for Evaluation:** To rigorously test the performance and scalability of the CSAM architecture, we will generate a large, synthetic corpus of simulated player interactions. This will allow us to programmatically generate tens of thousands of "memories" and run controlled stress tests, measuring key quantitative metrics like retrieval time and memory usage as the memory archive grows.

5. Work Plan for Next Semester

Month	Milestone
December 2025	Finalize project architecture. Set up Godot project with C++ GDExtension boilerplate.
January 2026	Implement core C++ memory structures (L2/L3 stores). Begin LLM fine-tuning.
February 2026	Implement the asynchronous Continuous Learning (summarization) pipeline.
February 2026	Integrate HNSWLib and implement the O(log N) SAMRetriever in C++.
March 2026	Implement the MMR diversity mechanism. Connect the full retrieval pipeline to the NPC controller.
March 2026	Develop the frontend UI for the demonstration (dialogue logs, performance dashboard).
April 2026	Conduct quantitative stress tests, record results, and prepare the final project demo and report.

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